

DDA: Dynamic Device Allocation — Latency Pool + Throughput Pool

Llama2-13B | 40 Decoder Layers | 20 total CXL devices split: N_L=10 + N_T=10

[L] Pool L — Latency Pool

N_L = 10 devices | TP = 10 | PP = 1

↓ Interactive Request (SLO = 1s)



Broadcast input → parallel GEMV (each handles 1/10 FC weights) → All-Reduce

PP = 1: All 40 Decoder Layers processed in ONE stage

Layers 1 - 40 (all in one stage, processed by TP group)

1 token in-flight at a time (PP=1, no pipeline)



TTFT = 0.882 s ✓ Meets 1s SLO

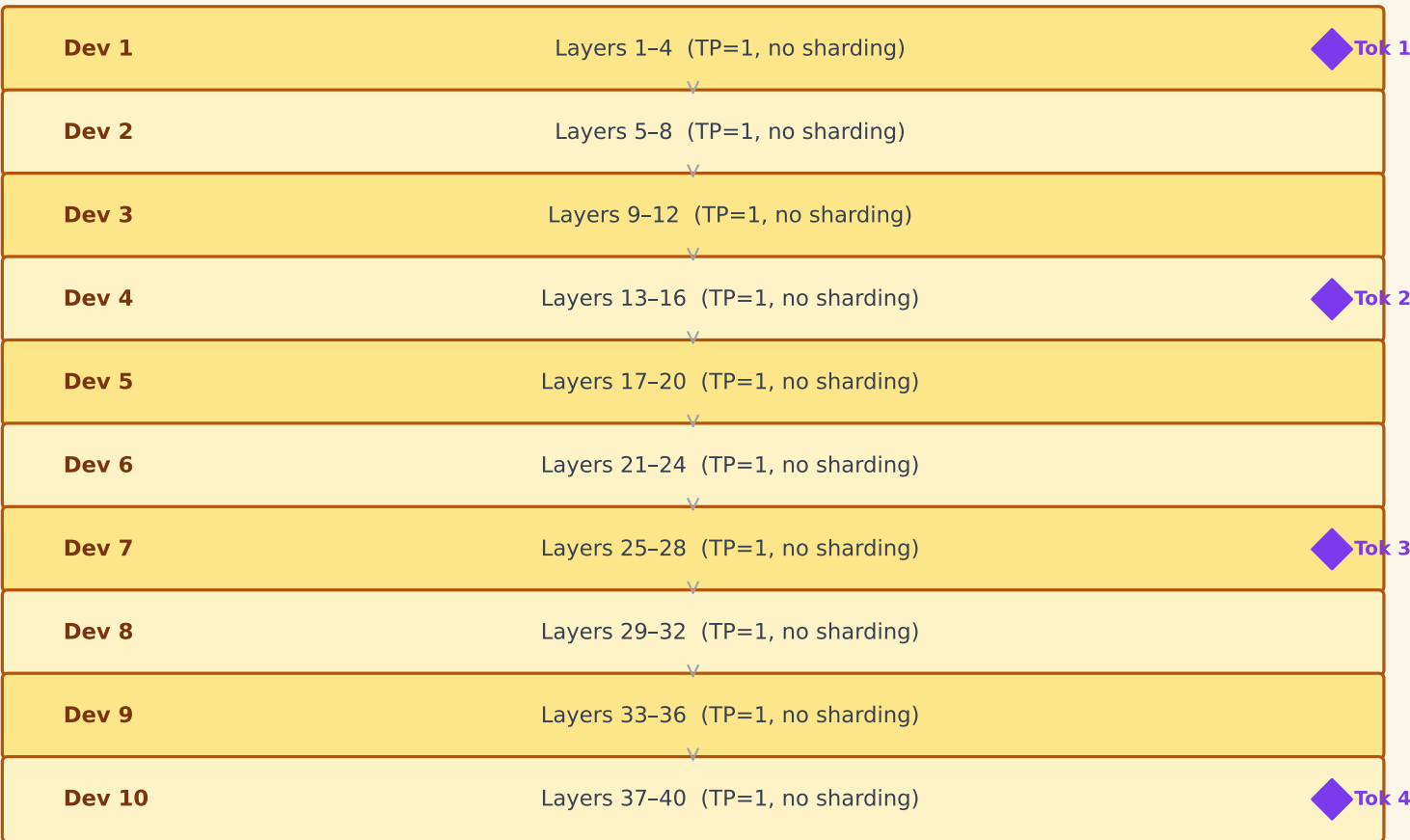
Pool L tput = 390 tok/s (low — single-token throughput)

High TP → each token uses all 10 devices → fastest per-token latency

[T] Pool T — Throughput Pool

N_T = 10 devices | TP = 1 | PP = 40

↓ Batch Requests (high-throughput workload)



4 tokens in-flight simultaneously (pipeline parallelism)

Pool T tput = 2116 tok/s (high pipeline throughput)

TTFT >> 1s (batch/offline only — not SLO-critical)

High PP → many tokens in-flight → maximizes device utilization

DDA System Throughput = Pool_L_tput + Pool_T_tput = 390 + 2116 = 2506 tok/s vs Best static (PP=2,TP=10): 780 tok/s → 3.2× gain